

Internship Report

Real Stable Polynomials and their Applications

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June 4, 2026

Introduction

These notes were written out of a series of presentations I gave at IMSc during the summer of 2025, while interning under Dr. Vikram Sharma's group. I'm truly grateful to Dr. Vikram Sharma and the entire group for their patience in sitting through my often messy and disorganized talks, and for all the kind feedback that helped me improve so much. Their enthusiasm really motivated me to explore Theoretical Computer Science more deeply — it was a wonderful summer!

We begin with the unimodality of Poisson binomial distributions, then study matching polynomials and prove the real rootedness of the matching-defect polynomial (Heilmann-Lieb). We discuss the permanent and its applications, including Gurvits's proof of van der Waerden's conjecture. Finally, we present the classical Lee-Yang theorem on phase transitions and characterisation of Lee-Yang stability preserving operators.

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1 Poisson Binomial Distributions

Definition 1.1 (Poisson Binomial Distribution). Let $X_1, X_2, \dots, X_n$ be independent Bernoulli random variables with parameters $b_1, b_2, \dots, b_n$, respectively; that is, $X_i \sim \text{Bern}(b_i)$ for each $i = 1, \dots, n$. Then the sum

$$X = \sum_{i=1}^n X_i$$

is said to follow a **Poisson binomial distribution**.

We aim to prove that the probability mass function of X is **unimodal**. That is, there exists an index $k \in \{1, \dots, n\}$ such that

$$p_1 \leq p_2 \leq \dots \leq p_k \geq p_{k+1} \geq \dots \geq p_n,$$

where $p_k = \mathbb{P}(X = k)$.

Theorem 1.2 (Newton's Inequalities). *Let $f(x) = \sum_{k=0}^n a_k x^k$ be a real polynomial of degree n with all real roots and non-negative coefficients $a_k \geq 0$. Then the coefficients satisfy*

$$\left(\frac{a_k}{\binom{n}{k}} \right)^2 \geq \left(\frac{a_{k-1}}{\binom{n}{k-1}} \right) \left(\frac{a_{k+1}}{\binom{n}{k+1}} \right) \quad \text{for } 1 \leq k \leq n-1.$$

Now consider the Poisson binomial random variable $X = \sum_{i=1}^n X_i$, where $X_i \sim \text{Bern}(b_i)$. The probability generating function (p.g.f) of X is given by:

$$q(x) := \sum_{k=0}^n p_k x^k = \prod_{i=1}^n (b_i x + 1 - b_i).$$

This is a product of linear polynomials with real coefficients, so $q(x)$ has all real roots. In fact, the roots are at

$$x_i = \frac{1 - b_i}{b_i}, \quad \text{since } b_i \in (0, 1).$$

Hence, $q(x)$ is a real-rooted polynomial with non-negative coefficients $p_k \geq 0$. We can now apply Newton's inequalities.

It follows that

$$\left(\frac{p_k}{\binom{n}{k}} \right)^2 \geq \left(\frac{p_{k-1}}{\binom{n}{k-1}} \right) \left(\frac{p_{k+1}}{\binom{n}{k+1}} \right) \quad \text{for } 1 \leq k \leq n-1.$$

Multiplying both sides by the binomial coefficients yields:

$$p_k^2 \geq p_{k-1} p_{k+1},$$

which shows that the sequence $\{p_k\}_{k=0}^n$ is **log-concave**.

It is a known fact that any log-concave sequence of nonnegative real numbers is unimodal. Hence, the pmf of the Poisson binomial distribution is unimodal.

We now ask the following question: given a real-rooted polynomial

$$p(x) = \sum_{i=0}^n a_i x^i, \quad \text{where } p(1) = 1,$$

do there exist independent random variables $X_i \sim \text{Bern}(b_i)$, with $b_i \in (0, 1)$, such that

$$X = \sum_{i=1}^n X_i$$

and the probability generating function (pgf) of X is equal to $p(x)$?

Turns out the answer is true, and can be proved in a similar manner.

2 Matching defect polynomial and real rootedness

Consider the graph $G = (V, E)$ with $w : E \mapsto \mathbb{R}$ as the weight function. Define,

$$m_k = \sum_{|M|=k} \prod_{e \in M} w_e \quad \text{where } M \text{ is a matching.}$$

Definition 2.1 (Matching defect polynomial). The matching defect polynomial of a graph can be defined as

$$\mu_G(x) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^k m_k x^{n-2k}.$$

Example 2.2. Consider the path graph on n vertices, denoted by P_n , where all edge weights are equal to 1. In this case, the coefficient m_k simply counts the number of matchings with k edges.

It is known that the number of k -edge matchings in P_n is given by:

$$m_k = \binom{n-k}{k}.$$

Therefore, the matching polynomial of P_n is:

$$\mu_{P_n}(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} x^{n-2k}.$$

Theorem 2.3. The matching polynomial $\mu_G(x)$ satisfies the recurrence:

$$\mu_G(x) = x \mu_{G \setminus \{v\}}(x) - \sum_{u \in N(v)} w_{uv} \mu_{G \setminus \{u, v\}}(x) \quad \text{where } \mu_\emptyset(x) = 1.$$

Proof. We compare the coefficients of x^{n-2k} on both sides.

Let m_k be the total weight of k -edge matchings in G , m'_k be the total weight of k -edge matchings in $G \setminus \{v\}$, $m''_{k-1}(u)$ be the total weight of $(k-1)$ -edge matchings in $G \setminus \{u, v\}$.

Then:

$$(-1)^k m_k = (-1)^k m'_k - \sum_{u \in N(v)} (-1)^k w_{uv} \cdot m''_{k-1}(u).$$

Since both terms contribute to x^{n-2k} , their signs match. Therefore,

$$m_k = m'_k + \sum_{u \in N(v)} w_{uv} \cdot m''_{k-1}(u),$$

which justifies the recurrence for all k , hence for the full polynomial. $\square$

Theorem 2.4 (Heilmann–Lieb '72). *Given a graph $G = (V, E)$, with weight function $w : E \mapsto \mathbb{R}_{\geq 0}$, the matching defect polynomial $\mu_G(x)$ is real-rooted.*

Definition 2.5. Let $p(x) = C_1 \prod_{i=1}^n (x - \lambda_i)$ and $q(x) = C_2 \prod_{i=1}^m (x - \nu_i)$ be two polynomials such that their degrees differ by at most 1. W.l.o.g. assume $n = \deg(p) \geq \deg(q) = m$. If

$$\lambda_n \leq \nu_n \leq \dots \leq \nu_1 \leq \lambda_1 \quad \text{when } m = n,$$

$$\nu_{n-1} \leq \lambda_{n-1} \leq \dots \leq \nu_1 \leq \lambda_1 \quad \text{when } m = n - 1,$$

we say q interlaces p . If the inequalities are strict it becomes a **strict interlacing**.

We shall first prove this theorem for complete graphs (K_n) , with positive edge weights and then extend the argument.

Theorem 2.6. *Let $G = K_n$ be the complete graph on n vertices, with strictly positive edge weights $w_{uv} > 0$ for all $u, v \in V(G)$. Then the matching polynomial $\mu_G(x)$ is real-rooted, and for every $v \in V(G)$, the polynomial $\mu_{G \setminus \{v\}}(x)$ strictly interlaces $\mu_G(x)$.*

Proof. We proceed by induction on n .

Base Case. Let $G = K_2 = (\{u, v\}, \{\{u, v\}\})$ with edge weight $w_{uv} > 0$. Then:

$$\mu_G(x) = x^2 - w_{uv}, \quad \mu_{G \setminus \{v\}}(x) = x.$$

Clearly, $\mu_G(x)$ has two real roots $\pm \sqrt{w_{uv}}$, which strictly interlace the single root 0 of $\mu_{G \setminus \{v\}}(x)$. Hence, the base case holds.

Inductive Hypothesis. Assume that for every complete graph $H = K_{n-1}$ with positive edge weights:

- (i) $\mu_H(x)$ is real-rooted,
- (ii) for every $w \in V(H)$, the polynomial $\mu_{H \setminus \{w\}}(x)$ strictly interlaces $\mu_H(x)$.

Let $G = K_n$, and fix any vertex $v \in V(G)$. Let

$$\lambda_{n-1} < \lambda_{n-2} < \dots < \lambda_1$$

be the (real) roots of $\mu_{G \setminus \{v\}}(x)$, listed in decreasing order.

Now for each $u \in N(v) = V(G) \setminus \{v\}$, the graph $G \setminus \{u, v\} \cong K_{n-2}$ is also a weighted complete graph with positive weights. By the inductive hypothesis, $\mu_{G \setminus \{u, v\}}(x)$ is real-rooted and strictly interlaces $\mu_{G \setminus \{v\}}(x)$. In particular, each such $\mu_{G \setminus \{u, v\}}(x)$ has roots that lie between the λ_i 's, i.e., for each $i = 1, \dots, n-2$, there exists a root in $(\lambda_{i+1}, \lambda_i)$, and:

$$\text{sign}(\mu_{G \setminus \{u, v\}}(\lambda_i)) = (-1)^{i+1}.$$

Now, consider the recurrence:

$$\mu_G(x) = x \cdot \mu_{G \setminus \{v\}}(x) - \sum_{u \in N(v)} w_{uv} \cdot \mu_{G \setminus \{u,v\}}(x).$$

Evaluating at $x = \lambda_i$, note that:

$$\mu_{G \setminus \{v\}}(\lambda_i) = 0 \implies \mu_G(\lambda_i) = - \sum_{u \in N(v)} w_{uv} \cdot \mu_{G \setminus \{u,v\}}(\lambda_i).$$

Each term $\mu_{G \setminus \{u,v\}}(\lambda_i)$ has sign $(-1)^{i+1}$, and since the weights $w_{uv} > 0$, the full sum is positive when i is even, and negative when i is odd. Hence:

$$\text{sign}(\mu_G(\lambda_i)) = (-1)^i.$$

Thus, between each pair λ_{i+1}, λ_i , the function $\mu_G(x)$ changes sign, so by the intermediate value theorem, there exists a root $\alpha_i \in (\lambda_{i+1}, \lambda_i)$. Therefore, we have $n - 2$ real roots lying strictly between the λ_i 's.

To obtain the remaining two roots:

- Note that $\mu_G(x)$ is monic of degree n , and $\mu_G(x) \rightarrow \infty$ as $x \rightarrow \infty$.
- Since $\mu_G(\lambda_1) < 0$, there exists $\alpha_0 > \lambda_1$ such that $\mu_G(\alpha_0) = 0$.
- Similarly, as $x \rightarrow -\infty$, by sign change across λ_{n-1} , there exists $\alpha_n < \lambda_{n-1}$ such that $\mu_G(\alpha_n) = 0$.

Thus, $\mu_G(x)$ has n real roots $\alpha_n < \dots < \alpha_1 < \alpha_0$ that strictly interlace the λ_i 's. The theorem is proved by induction. $\square$

Lemma 2.7. *For a sequence of polynomials $f_1(z), f_2(z), \dots$ in the complex variable z , such that their degrees are uniformly bounded, let $\Omega \subset \mathbb{C}$ be an open subset, and $f_n \rightarrow f \in \mathbb{C}[z]$ coefficient-wise. If each f_m is Ω -stable then either f is Ω -stable or $f \equiv 0$.*

We shall use the above lemma to prove that the matching polynomial is **real-rooted** for any graph G .

Let $G(V, E)$ be a graph on n vertices with $w : E \rightarrow \mathbb{R}$. Let $E' = V \times V$. Define $G^{(n)}(V, E')$ with weight function $w' : E' \rightarrow \mathbb{R}$ defined as

$$w'(u, v) = \begin{cases} w_{uv} & \text{if } (u, v) \in E, \\ \frac{1}{n} & \text{if } (u, v) \notin E. \end{cases}$$

Clearly $\mu_{G^{(n)}}(x) \rightarrow \mu_G(x)$ as $n \rightarrow \infty$, and by our lemma $\mu_G(x)$ is real-rooted, since we have proved $\mu_{G^{(n)}}(x)$ to be real-rooted for all $n \in \mathbb{N}$.

Write $p(x) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} m_k x^k$. We can write $\mu_G(x) = x p(-x^2)$ for odd n , and $\mu_G(x) = p(-x^2)$ for even n . Since $\mu_G(x)$ is real-rooted, it follows that $p(x)$ must be real-rooted as well.

Hence $\{m_k\}_{k=0}^{\lfloor \frac{n}{2} \rfloor}$ is unimodal for all graphs $G(V, E)$.

3 Permanent and its applications

Given a matrix $A = [a_{ij}]$ be a $n \times n$ matrix, we define the permanent of A as,

$$\text{perm}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i,\sigma(i)}.$$

We notice the similarity with the definition of the determinant which is defined as,

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{\text{inv}(\sigma)} \prod_{i=1}^n a_{i,\sigma(i)},$$

where S_n is the symmetric group on n -elements and $\text{inv}(\sigma)$ denotes the number of inversions in the permutation σ .

While the structure of the definition might look similar, the absence of the inversion term in the permanent makes it lose a lot of the “clean” properties possessed by the determinant, such as being multiplicative and being computable in polynomial time.

We have some properties of the permanent that are useful to us.

Theorem 3.1. *For $k \in \mathbb{R}$, if A is a $n \times n$ matrix then*

$$\text{perm}(kA) = k^n \text{perm}(A).$$

Theorem 3.2. *If $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $n \times n$ matrices with non-negative real entries. Suppose*

$$a_{ij} \leq b_{ij} \quad \forall 1 \leq i, j \leq n.$$

Then we have

$$\text{perm}(A) \leq \text{perm}(B).$$

The permanent however is interesting to look at because of its combinatorial properties; we shall look at a few of them.

3.1 Counting Derangements

Suppose you have n letters and n envelopes corresponding to each letter. Suppose we want to shuffle the letters in such a way such that no letter goes to its original envelope. Now we pose the question, in how many ways is this possible?

The problem first proposed by Pierre Rémond de Montmort in 1708, has a very neat solution using the Principle of Inclusion and Exclusion, and the final expression we get is,

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!} \right),$$

where D_n denotes the number of such arrangements.

Definition 3.3 (Fixed point). Let $\sigma \in S_n$ be a permutation. We define $1 \leq i \leq n$ to be a *fixed point* if $\sigma(i) = i$.

Definition 3.4. Let $\sigma \in S_n$ be a permutation. We say σ is a *derangement* if σ has no fixed points.

The number of such configurations is clearly the number of derangements in S_n .

Theorem 3.5. Let $A = J - I$, where J is the all-ones matrix and I is the identity matrix. Then $\text{perm}(A) = D_n$.

Proof. Clearly,

$$\text{perm}(A) = \text{perm}(J - I) = \sum_{\sigma \in S_n} \prod_{i=1}^n (J - I)_{i, \sigma(i)}.$$

Since $(J - I)_{i,i} = 0$ for all $1 \leq i \leq n$, we have

$$\text{perm}(A) = \sum_{\substack{\sigma \in S_n \\ \sigma(i) \neq i \forall i}} 1 = D_n,$$

which proves our claim. $\square$

3.2 Bipartite Matchings

Definition 3.6 (Bipartite Graph). A graph $G(V \cup W, E)$ is said to be bipartite if for all $(u, v) \in E$, we have either $u \in V$ and $v \in W$, or vice versa, with $V \cap W = \emptyset$.

Definition 3.7 (Matching). $M \subseteq E$ is a matching if no two edges in M share a vertex.

Given a bipartite graph $G(V \cup W, E)$, it is clear a perfect matching only exists when $|V| = |W| = n$. Write $V = \{u_1, u_2, \dots, u_n\}$ and $W = \{w_1, w_2, \dots, w_n\}$. Given the graph, define its biadjacency matrix $A_G = [a_{ij}]$ by

$$a_{ij} = \begin{cases} 1 & \text{if } (u_i, w_j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 3.8. The number of perfect matchings in G is equal to the permanent of the biadjacency matrix, i.e., $\text{perm}(A_G) = \#\text{perfect matchings in } G$.

Proof. Let $G(V \cup W, E)$ be a bipartite graph with $V \cap W = \emptyset$, where $|V| = |W| = n$. Define the biadjacency matrix $A_G = [a_{ij}]$ of size $n \times n$ in the manner described above.

The permanent of A_G is given by

$$\text{perm}(A_G) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Each permutation σ corresponds to a potential perfect matching that pairs vertex v_i with vertex $w_{\sigma(i)}$.

- If any $a_{i, \sigma(i)} = 0$, then the edge $(v_i, w_{\sigma(i)})$ does not exist in G , so σ does not represent a valid perfect matching.
- If for all i , $a_{i, \sigma(i)} = 1$, the edges $\{(v_i, w_{\sigma(i)}) \mid i = 1, \dots, n\}$ form a perfect matching, since each vertex in V and W is matched exactly once.

Hence, the permanent counts exactly the number of perfect matchings in G . $\square$

3.3 van der Waerden's Conjecture

Definition 3.9. A matrix $A = [a_{ij}]$ of size $n \times n$ is said to be doubly stochastic if:

- All entries are non-negative: $a_{ij} \geq 0 \quad \forall 1 \leq i, j \leq n$.
- Each row sums to 1: $\sum_{j=1}^n a_{ij} = 1 \quad \forall 1 \leq i \leq n$.
- Each column sums to 1: $\sum_{i=1}^n a_{ij} = 1 \quad \forall 1 \leq j \leq n$.

While studying the properties of doubly stochastic matrices, Bartel Leendert van der Waerden in 1926 made a conjecture about the extreme values of these matrices. It is particularly interesting because permanents, unlike determinants, are notoriously difficult to analyse.

Conjecture (van der Waerden). For any $n \times n$ doubly stochastic matrix A , the permanent satisfies

$$\text{perm}(A) \geq \frac{n!}{n^n},$$

with equality if and only if A is the matrix with all entries $\frac{1}{n}$.

This problem remained open for decades until Egorychev and Falikman proved it in 1981.

Later we shall see a proof of this theorem using the theory of real stable polynomials due to Gurvits.

Earlier we saw that $\mathbf{perm}(J - I) = D_n$. Notice that the sum of each row and column in $J - I$ is $n - 1$. Hence $\frac{1}{n-1}(J - I)$ is a doubly stochastic matrix. And hence we have by van der Waerden's theorem,

$$\mathbf{perm}\left(\frac{1}{n-1}(J - I)\right) \geq \frac{n!}{n^n} \implies D_n \geq n! \left(1 - \frac{1}{n}\right)^n.$$

And as $n \rightarrow \infty$ we have,

$$D_n \gtrsim \frac{n!}{e}.$$

And thus we have a useful lower bound.

The same line of reasoning can be used to argue that,

$$\#\text{perfect matchings in } k\text{-regular bipartite graphs} \geq \frac{n! k^n}{n^n} \geq \left(\frac{k}{e}\right)^n.$$

4 Real Stable Polynomials

Definition 4.1. $\mathcal{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$.

Definition 4.2 (Real Stability). A polynomial $p(z) \in \mathbb{C}[z_1, z_2, \dots, z_n]$ is said to be stable if it has no roots in $\mathcal{H}^n$. It is called real stable if all of its coefficients are real.

Theorem 4.3. A univariate polynomial $p(x) \in \mathbb{R}[x]$ is real-rooted if and only if it is real stable.

Proof. ($\Rightarrow$) Say $p(x)$ is real-rooted, then by definition it has no roots in $\mathcal{H}$ and hence it is real stable.

($\Leftarrow$) Suppose $p(x)$ is a real stable polynomial with a complex root $z \in \mathbb{C}$ such that $\text{Im}(z) < 0$. Then by the conjugate root theorem it must have a root $\bar{z}$ such that $\text{Im}(\bar{z}) > 0$, which violates our real stability condition, and thus there cannot be non-real roots. $\square$

Lemma 4.4. A multivariate polynomial $f(z_1, \dots, z_n) \in \mathbb{R}[z_1, \dots, z_n]$ is stable if and only if for every $v \in \mathbb{R}^n$ and every $u \in \mathbb{R}_{>0}^n$, the univariate polynomial $f(v + tu)$ is real-rooted.

Proof. Assume that for all $v \in \mathbb{R}^n$ and $u \in \mathbb{R}_{>0}^n$, the polynomial $f(v + tu)$ is real-rooted. Suppose, for the sake of contradiction, that f is not stable.

Then there exists a point $a = (a_1, \dots, a_n) \in \mathcal{H}^n$ such that $f(a) = 0$, where $\mathcal{H}$ denotes the open upper half-plane. Set $v = \text{Re}(a)$ and $u = \text{Im}(a)$. Since $a \in \mathcal{H}^n$, we have $u_i > 0$ for each i .

Now, observe that

$$f(a) = f(\text{Re}(a) + i \text{Im}(a)) = f(v + iu) = 0,$$

which means that $t = i$ is a root of the univariate polynomial $f(v + tu)$. However, this contradicts the assumption that $f(v + tu)$ is real-rooted, since i is not a real number.

Therefore, our assumption must be false, and f is stable if and only if every such univariate specialization is real-rooted. $\square$

4.1 Linear stability preserving Transformations

Theorem 4.5 (Borceä–Bränden '09). Let $\mathbb{C}_k[z_1, \dots, z_n]$ be the set of all polynomials such that z_j has degree at most k . Let $T : \mathbb{C}_k[z_1, \dots, z_n] \rightarrow \mathbb{C}[z_1, \dots, z_n]$ be a linear transformation with $\dim(\text{Im}(T)) > 2$. We define the symbol of T to be the polynomial

$$G_T(z_1, \dots, z_n, w_1, \dots, w_n) = T\left(\prod_{i=1}^n (z_i + w_i)^k\right),$$

where we treat each z_j as a variable while treating the w_i 's as constants. Then

$$G_T \text{ is real stable} \iff T \text{ is stability preserving.}$$

We shall use this classification to prove a couple of closure properties related to real stable polynomials.

4.2 Closure Properties

- **Product.** Let f, g be two real stable polynomials. Then their product $f \cdot g$ is also a real stable polynomial.
- **Specialisation.** Consider the transformation $f(z_1, \dots, z_n) \mapsto f(a, z_2, \dots, z_n)$ for $a \in \mathcal{H} \cup \mathbb{R}$; this is a linear transformation. We shall show this using the previous theorem.

Let k be the highest degree of any variable in f , so $f \in \mathbb{R}_k[z_1, \dots, z_n]$. For $a \in \mathcal{H} \cup \mathbb{R}$, define $T_a : \mathbb{R}_k[z_1, \dots, z_n] \rightarrow \mathbb{R}[z_1, \dots, z_n]$ by $T_a(f(z_1, \dots, z_n)) = f(a, z_2, \dots, z_n)$.

We compute the symbol of T_a :

$$G_{T_a}(z_1, \dots, z_n, w_1, \dots, w_n) = T_a \left(\prod_{i=1}^n (z_i + w_i)^k \right) = (a + w_1)^k \prod_{i=2}^n (z_i + w_i)^k.$$

Clearly G_{T_a} is real stable when $a \in \mathcal{H} \cup \mathbb{R}$, and hence T_a is stability preserving by Borceä–Bränden.

- **Differentiation.** Let $f \in \mathbb{R}_k[z_1, \dots, z_n]$ be a multivariate polynomial, and let $T : \mathbb{R}_k[z_1, \dots, z_n] \rightarrow \mathbb{R}[z_1, \dots, z_n]$ be $T(f) = \frac{\partial f}{\partial z_1}$. Clearly T is a linear operator. We have

$$G_T(z_1, \dots, z_n, w_1, \dots, w_n) = T \left(\prod_{i=1}^n (z_i + w_i)^k \right) = k(z_1 + w_1)^{k-1} \prod_{i=2}^n (z_i + w_i)^k.$$

Clearly G_T is real stable and hence T is stability preserving.

4.3 Yet another proof of Heilmann–Lieb

Let $G(V, E)$ be our graph with weight function $w : E \rightarrow \mathbb{R}_{\geq 0}$. Define

$$Q(z_v : v \in V) = \prod_{(u,v) \in E} (1 - w_{uv} z_u z_v).$$

We claim that Q is real stable, since $z_u z_v = \frac{1}{w_{uv}} > 0$, and the product of two numbers in $\mathcal{H}$ can never be a positive real.

Define the multi-affine part operator (MAP) as the linear operator that preserves all the multilinear monomials and eliminates everything else:

$$\text{MAP} \left(\prod_{i=1}^n z_i^{e_i} \right) = \begin{cases} \prod_{i=1}^n z_i^{e_i} & \text{if } e_i = 1 \ \forall \ 1 \leq i \leq n, \\ 0 & \text{otherwise.} \end{cases}$$

We claim that MAP is stability preserving. Consider $\text{MAP} : \mathbb{R}_k[z_1, \dots, z_n] \rightarrow \mathbb{R}[z_1, \dots, z_n]$. Its symbol is

$$G_{\text{MAP}}(z_1, \dots, z_n, w_1, \dots, w_n) = \text{MAP} \left(\prod_{i=1}^n (z_i + w_i)^k \right) = \prod_{i=1}^n (w_i^k + k z_i w_i^{k-1}) = \prod_{i=1}^n w_i^{k-1} (w_i + k z_i).$$

Hence G_{MAP} is real stable and MAP is stability preserving.

Now, since each monomial of Q is a collection of edges of the graph G , applying the MAP operator to Q gives us the matchings of G . Hence,

$$\text{MAP}(Q) = \sum_{\text{matching } M} (-1)^{|M|} \prod_{(u,v) \in M} w_{uv} z_u z_v$$

is real stable. This can be converted to a real-rooted polynomial by diagonalisation, and then inversion and multiplication by x^n gives us $\mu_G(x)$. Since each of these transformations is real-stability-preserving, $\mu_G(x)$ is real-rooted.

Which eventually leads us to the fact that $\{m_k\}_{k=0}^{\lfloor \frac{n}{2} \rfloor}$ is unimodal.

A related combinatorial problem that we can analyze using the theory of graph matchings is the classical problem of *rook placements* on an $n \times n$ chessboard.

Let r_k denote the number of ways to place k non-attacking rooks on an $n \times n$ chessboard. We claim that the sequence $(r_0, r_1, \dots, r_n)$ is *unimodal*.

Proof. Consider the complete bipartite graph $K_{n,n}$, where the two partite sets represent the rows and columns of the chessboard, respectively. Each edge in $K_{n,n}$ corresponds to a cell on the board, determined by a unique row and column pair.

A *rook placement* corresponds to selecting a subset of cells such that no two rooks are in the same row or column. Equivalently, this is the same as selecting a *matching* in $K_{n,n}$, where each edge in the matching represents a rook placed at a distinct row-column pair.

Thus, the number r_k of ways to place k non-attacking rooks on the board equals the number of *matchings of size k* in $K_{n,n}$. It is well-known that the sequence of matching numbers $m_k(K_{n,n})$ is *log-concave*, and hence *unimodal*.

Therefore, the sequence $(r_0, r_1, \dots, r_n)$, counting rook placements, is unimodal. $\square$

5 van der Waerden's Theorem

In this section we present Gurvits's proof of the van der Waerden conjecture using the theory of real stable polynomials and the notion of *capacity* of a polynomial.

5.1 Setup

Let $A = [a_{ij}]$ be an $n \times n$ doubly stochastic matrix. Define the multilinear polynomial

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} x_j \right).$$

Each factor $\sum_{j=1}^n a_{ij} x_j$ is a linear form in $x_1, \dots, x_n$ with non-negative coefficients. The coefficient of the monomial $x_1 x_2 \cdots x_n$ in p is precisely

$$[x_1 x_2 \cdots x_n] p = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)} = \mathbf{perm}(A).$$

Since p is a product of linear forms with non-negative real coefficients, it is a real stable polynomial (each linear factor has no zeros in $\mathcal{H}^n$, and products preserve stability). Moreover, partial differentiation is stability-preserving (as shown in the previous section), so

$$\mathbf{perm}(A) = \frac{\partial^n p}{\partial x_1 \partial x_2 \cdots \partial x_n} \Big|_{x=0}.$$

5.2 Capacity

Definition 5.1 (Capacity). For a polynomial $p(x_1, \dots, x_n)$ with non-negative coefficients, define its *capacity* as

$$\text{cap}(p) = \inf_{x_1, \dots, x_n > 0} \frac{p(x_1, x_2, \dots, x_n)}{x_1 x_2 \cdots x_n}.$$

5.3 The Capacity Lemmas

Lemma 5.2. For any univariate real-rooted polynomial $R(x)$ of degree k with non-negative coefficients,

$$\text{cap}(R) \leq \left(\frac{k}{k-1} \right)^{k-1} R'(0)$$

for $k \geq 2$, and $\text{cap}(R) = R'(0)$ for $k = 1$.

Proof. Since R is real-rooted with non-negative coefficients, all of its roots lie on the non-positive real axis.

Case $k = 1$. Write $R(x) = ax + b$. Then

$$\text{cap}(R) = \inf_{x > 0} \frac{ax + b}{x} = \lim_{x \rightarrow \infty} \frac{ax + b}{x} = a = R'(0).$$

Case $k \geq 2$, $R(0) = 0$. At least one root $\alpha_i = 0$, so $R(x) = x \cdot \tilde{R}(x)$ where $\tilde{R}$ has degree $k - 1$ with non-negative roots. Then

$$\text{cap}(R) = \inf_{x>0} \tilde{R}(x) = \tilde{R}(0) = 0,$$

and the bound holds trivially since $R'(0) \geq 0$.

Case $k \geq 2$, $R(0) > 0$. All roots are strictly negative; write $R(x) = \prod_{i=1}^k (x + \alpha_i)$ with $\alpha_i > 0$. Set $a_i = 1/\alpha_i > 0$. Then

$$\frac{R(x)}{x} = \frac{1}{x} \prod_{i=1}^k \frac{1}{a_i} \prod_{i=1}^k (a_i x + 1).$$

By the AM-GM inequality applied to the product:

$$\prod_{i=1}^k (a_i x + 1) \leq \left(\frac{1}{k} \sum_{i=1}^k (a_i x + 1) \right)^k = \left(1 + \frac{x}{k} \sum_{i=1}^k a_i \right)^k.$$

Setting $S = \sum_{i=1}^k a_i$ and optimising over $x > 0$ by choosing $x^* = \frac{k}{(k-1)S}$ gives

$$\frac{R(x^*)}{x^*} \leq \frac{S}{\prod_{i=1}^k a_i} \cdot \left(\frac{k}{k-1} \right)^{k-1} \cdot \frac{1}{k} \cdot \left(1 + \frac{1}{k-1} \right)^k.$$

One checks that $\frac{S}{\prod_{i=1}^k a_i} \cdot \frac{1}{k} \cdot \frac{(k/(k-1))^{k-1} (k/(k-1))}{1} = \left(\frac{k}{k-1} \right)^{k-1} R'(0)$, since

$$R'(0) = \sum_{i=1}^k \prod_{j \neq i} \alpha_j = \frac{\sum_{i=1}^k a_i}{\prod_{i=1}^k a_i} \cdot \frac{1}{k} \cdot k = \frac{S}{\prod_{i=1}^k a_i}.$$

Hence

$$\text{cap}(R) \leq \left(\frac{k}{k-1} \right)^{k-1} R'(0). \quad \square$$

Lemma 5.3. *For any stable polynomial $p(x_1, x_2, \dots, x_n)$ of degree n with non-negative coefficients,*

$$\text{cap}(p) \leq \frac{n^n}{n!} \frac{\partial^n p}{\partial x_1 \partial x_2 \cdots \partial x_n} \Big|_{x=0}.$$

Proof. We proceed by induction on n .

Base case $n = 1$. Here $p(x_1)$ is a univariate stable polynomial, hence real-rooted. By the univariate lemma (Lemma 5.2),

$$\text{cap}(p) = p'(0) = \frac{1^1}{1!} p'(0),$$

which is exactly the claimed bound.

Inductive step. Suppose the bound holds for all stable polynomials of degree $n - 1$. Let p be stable of degree n . Define the *partial derivative at zero* in x_1 :

$$r(x_2, \dots, x_n) = \frac{\partial p}{\partial x_1} \Big|_{x_1=0}.$$

Since p has degree at most n in x_1 and p is stable, r is a stable polynomial of degree $n - 1$ in $(x_2, \dots, x_n)$ with non-negative coefficients (differentiation and specialisation both preserve stability).

By the univariate capacity lemma applied to $x_1 \mapsto p(x_1, x_2, \dots, x_n)$ for fixed $(x_2, \dots, x_n) > 0$ (which is real-rooted of degree n in x_1):

$$\inf_{x_1 > 0} \frac{p(x_1, x_2, \dots, x_n)}{x_1} \leq \left(\frac{n}{n-1} \right)^{n-1} r(x_2, \dots, x_n).$$

Dividing both sides by $x_2 \cdots x_n > 0$ and taking the infimum over $(x_2, \dots, x_n) > 0$:

$$\text{cap}(p) \leq \left(\frac{n}{n-1} \right)^{n-1} \text{cap}(r).$$

Applying the inductive hypothesis to r (degree $n - 1$, stable):

$$\text{cap}(r) \leq \frac{(n-1)^{n-1}}{(n-1)!} \frac{\partial^{n-1} r}{\partial x_2 \cdots \partial x_n} \Big|_{x_2=\dots=x_n=0}.$$

Since $\frac{\partial^{n-1} r}{\partial x_2 \cdots \partial x_n} \Big|_0 = \frac{\partial^n p}{\partial x_1 \cdots \partial x_n} \Big|_0$, combining gives

$$\text{cap}(p) \leq \left(\frac{n}{n-1} \right)^{n-1} \cdot \frac{(n-1)^{n-1}}{(n-1)!} \cdot \frac{\partial^n p}{\partial x_1 \cdots \partial x_n} \Big|_0 = \frac{n^n}{n!} \frac{\partial^n p}{\partial x_1 \cdots \partial x_n} \Big|_0. \quad \square$$

5.4 Proof of van der Waerden's Theorem

We now have all the ingredients to prove the conjecture.

Theorem 5.4 (van der Waerden). *Let $A = [a_{ij}]$ be an $n \times n$ doubly stochastic matrix. Then*

$$\mathbf{perm}(A) \geq \frac{n!}{n^n},$$

with equality if and only if $A = J_n$, the $n \times n$ all- $\frac{1}{n}$ matrix.

Proof. Let $p(x_1, \dots, x_n) = \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} x_j \right)$ as defined above. We have established that p is stable with non-negative coefficients and that $\mathbf{perm}(A) = \frac{\partial^n p}{\partial x_1 \cdots \partial x_n} \Big|_{x=0}$.

Step 1: Lower bound on capacity. For any $x_1, \dots, x_n > 0$, use the AM-GM inequality on each row factor: since $\sum_j a_{ij} = 1$,

$$\sum_{j=1}^n a_{ij} x_j \geq \prod_{j=1}^n x_j^{a_{ij}}.$$

Therefore,

$$p(x_1, \dots, x_n) \geq \prod_{i=1}^n \prod_{j=1}^n x_j^{a_{ij}} = \prod_{j=1}^n x_j^{\sum_{i=1}^n a_{ij}} = \prod_{j=1}^n x_j,$$

where the last equality uses the column-stochasticity condition $\sum_i a_{ij} = 1$. Hence

$$\frac{p(x_1, \dots, x_n)}{x_1 x_2 \cdots x_n} \geq 1 \quad \text{for all } x > 0,$$

which gives

$$\text{cap}(p) \geq 1.$$

Step 2: Conclude. By Lemma 5.3,

$$1 \leq \text{cap}(p) \leq \frac{n^n}{n!} \mathbf{perm}(A),$$

and rearranging yields the desired bound:

$$\mathbf{perm}(A) \geq \frac{n!}{n^n}.$$

Equality. Equality holds if and only if $\text{cap}(p) = 1$ and the AM-GM bound is tight in every row, which requires $a_{ij}x_j$ to be equal for all j in each row. Since this must hold at the optimal $x = (1, 1, \dots, 1)$, one deduces $a_{ij} = \frac{1}{n}$ for all i, j , i.e., $A = J_n$. $\square$

6 Lee-Yang Theory of Phase Transitions

Consider a graph $G = (V, E)$ with an associated configuration space $\Omega = \Omega(G)$, whose elements $\sigma \in \Omega$ represent combinatorial structures defined over G — such as spin assignments, matchings, or colorings. We equip Ω with a weight function $w : \Omega \rightarrow \mathbb{R}_{>0}$, which induces a probability distribution

$$\mathbb{P}(\sigma) = \frac{w(\sigma)}{Z}, \quad Z := \sum_{\sigma \in \Omega} w(\sigma).$$

Beyond estimating Z itself, one is often interested in the expected value of an *observable* $f : \Omega \rightarrow \mathbb{R}_{\geq 0}$. Its expectation under $\mathbb{P}$ is

$$\langle f \rangle := \sum_{\sigma \in \Omega} \mathbb{P}(\sigma) f(\sigma) = \frac{\sum_{\sigma \in \Omega} w(\sigma) f(\sigma)}{Z}.$$

6.1 The Ferromagnetic Ising Model

A usual use of this framework is the *ferromagnetic Ising model*. Let $G = (V, E)$ be a graph. A configuration is a spin assignment $\sigma : V \rightarrow \{+1, -1\}$, so the configuration space is $\Omega = \{+1, -1\}^V$ with $|\Omega| = 2^{|V|}$. The model is parametrised by an *edge parameter* $\beta \in (0, 1)$, and a *vertex activity* $\lambda > 0$. For a configuration σ , define

$$d(\sigma) := |\{ \{u, v\} \in E : \sigma(u) \neq \sigma(v) \}|, \quad m(\sigma) := |\{v \in V : \sigma(v) = +1\}|,$$

as the number of *cut edges* and the number of +1-spin vertices respectively. The weight is then

$$w(\sigma) := \beta^{d(\sigma)} \lambda^{m(\sigma)},$$

and the associated partition function is

$$Z_\beta(\lambda) := \sum_{\sigma \in \{+1, -1\}^V} w(\sigma) = \sum_{\sigma \in \{+1, -1\}^V} \beta^{d(\sigma)} \lambda^{m(\sigma)}.$$

Since $m(\sigma) \in \{0, 1, \dots, |V|\}$, fixing β and viewing Z as a function of λ yields a univariate polynomial of degree at most $|V|$:

$$Z(\lambda) = \sum_{k=0}^{|V|} c_k(\beta) \lambda^k.$$

The *magnetization* $m(\sigma)$ counts the number of +1-spins, with mean

$$M_\beta(\lambda) = \mathbb{E} \left(\frac{m(\sigma)}{|V|} \right) = \frac{1}{|V|} \sum_{\sigma \in \{-1, +1\}^{|V|}} \frac{m(\sigma) \beta^{d(\sigma)} \lambda^{m(\sigma)}}{Z_\beta(\lambda)}.$$

6.2 Free Energy and Phase Transitions

The *free energy* per unit volume of the model is defined as

$$F_\beta(\lambda) := -\frac{1}{|V|} \log Z_\beta(\lambda).$$

This quantity plays a central role in the study of phase transitions. We can write our magnetisation $M_\beta(\lambda)$ in terms of the free energy $F_\beta(\lambda)$,

$$M_\beta(\lambda) := \frac{1}{|V|} \frac{\sum_{\sigma \in \{+, -\}^V} m(\sigma) \beta^{d(\sigma)} \lambda^{m(\sigma)}}{Z_\beta(\lambda)} = \frac{1}{|V| Z_\beta(\lambda)} \cdot z \frac{\partial}{\partial z} Z_\beta(z) \Big|_{z=\lambda} = -z \frac{\partial F_\beta(z)}{\partial z} \Big|_{z=\lambda}.$$

A *phase transition* of a physical system is defined as a discontinuity or singularity of the observable being considered (in our case, the quantity $M_\beta(\lambda)$), or of one of its derivatives.

We therefore see that:

1. If the free energy $F(z)$ is an analytic function for $z \in D$, then, by definition, the system *cannot* have a phase transition in the region D of the parameter space.
2. Since the only “physically” interesting region of parameter is $\lambda > 0$, and the free energy of any *finite* graph is clearly an analytic function for $\lambda > 0$, *neither* of the above systems can exhibit a phase transition on a *finite* graph. To see this explicitly, recall that for a finite graph $G = (V, E)$, the partition function is a polynomial in λ of degree at most $|V|$,

$$Z_\beta(\lambda) = \sum_{k=0}^{|V|} c_k(\beta) \lambda^k,$$

with strictly positive coefficients $c_k(\beta) > 0$ for all $\beta \in (0, 1)$. Therefore, for $\lambda > 0$, we have $Z_\beta(\lambda) > 0$, so $\log Z_\beta(\lambda)$ is well-defined and smooth, and the free energy

$$F_\beta(\lambda) = -\frac{1}{|V|} \log Z_\beta(\lambda)$$

is analytic on all of $(0, \infty)$.

6.3 The Lee–Yang Theorem

To study where $Z_\beta(\lambda)$ can vanish, it is natural to pass to a multivariate generalisation, assigning an independent activity $\lambda_v > 0$ to each vertex $v \in V$.

Definition 6.1 (Multivariate partition function). Let $G = (V, E)$ be a graph on n vertices with $V = \{v_1, \dots, v_n\}$. The *multivariate partition function* is

$$Z_\beta(\lambda_1, \dots, \lambda_n) := \sum_{\sigma \in \{+1, -1\}^V} \beta^{d(\sigma)} \prod_{v: \sigma(v)=+1} \lambda_v.$$

We say G has the *Lee–Yang property* if, for every induced subgraph $H \subseteq G$,

$$|\lambda_v| > 1 \text{ for all } v \in V(H) \implies Z_\beta^H(\lambda) \neq 0.$$

Theorem 6.2 (Lee–Yang). *Let G be any finite graph and $\beta \in (0, 1)$. Then $Z_\beta(\lambda) \neq 0$ for all $\lambda \in \mathbb{C}$ with $|\lambda| \notin \{1\}$.*

Setting all $\lambda_v = \lambda$ in Definition 1 recovers $Z_\beta(\lambda)$, so Theorem 6.2 follows once we establish $Z_\beta(\lambda_1, \dots, \lambda_n) \neq 0$ for $|\lambda_v| > 1$. The case $|\lambda| < 1$ then follows from the fact

$$Z_\beta\left(\frac{1}{\lambda}\right) = \frac{1}{\lambda^n} Z_\beta(\lambda),$$

which shows that a zero at λ inside the unit disk would force a zero at $1/\lambda$ outside it, a contradiction.

The proof rests on the following contraction lemma.

Lemma 6.3 (Asano contraction). *Let H be a graph with the Lee–Yang property, and let $v_1, v_2 \in V(H)$ be non-adjacent. Then the graph H' obtained by merging v_1 and v_2 into a single vertex v also has the Lee–Yang property.*

Proof. Write the partition function of H , collecting terms by the spins at v_1 and v_2 , as

$$Z_\beta^H(\lambda_1, \lambda_2, \boldsymbol{\lambda}) = A\lambda_1\lambda_2 + B\lambda_1 + C\lambda_2 + D,$$

where A, B, C, D are polynomials in the remaining activities $\boldsymbol{\lambda} = (\lambda_3, \dots, \lambda_n)$. Since v_1 and v_2 share no edge, merging them identifies $\lambda_1 = \lambda_2 = \lambda$, and the mixed term $B\lambda_1 + C\lambda_2$ contributes $(B + C)\lambda$ to Z_β^H , while the contracted graph H' has partition function

$$Z_\beta^{H'}(\lambda, \boldsymbol{\lambda}) = A\lambda + D.$$

Fix $\boldsymbol{\lambda}$ with $|\lambda_i| > 1$ for all $i \geq 3$. We must show $A\lambda + D \neq 0$ for $|\lambda| > 1$. Suppose for contradiction that $|D/A| > 1$. By the Lee–Yang property of H , the polynomial $Ax^2 + (B + C)x + D$ (obtained by setting $\lambda_1 = \lambda_2 = x$) has no root with $|x| > 1$. Hence both roots lie in the closed unit disk, so their product satisfies $|D/A| \leq 1$, a contradiction. Therefore $|D/A| \leq 1$, which means $A\lambda + D = 0$ requires $|\lambda| \leq 1$. The argument extends similarly to induced subgraphs of H' . $\square$

Proof of Theorem 6.2. We induct on $|V|$.

Base cases. A single vertex has $Z_\beta(\lambda_1) = 1 + \lambda_1$, which is nonzero for $|\lambda_1| > 1$. A single edge has

$$Z_\beta(\lambda_1, \lambda_2) = 1 + \beta(\lambda_1 + \lambda_2) + \lambda_1\lambda_2,$$

and one verifies directly that $|\lambda_1|, |\lambda_2| > 1$ forces this to be nonzero, using the fact that the relevant Möbius transformation maps the exterior of the unit disk to its interior for $\beta < 1$.

Inductive step. We build G by starting from a disjoint union of $|E|$ edges and repeatedly applying Lemma 6.3. Concretely, each edge of G contributes one copy, giving a forest of $|E|$ disjoint edges; this has the Lee–Yang property since its partition function factors as a product over components. We then merge endpoints across components, one identification per vertex of G , until the graph is G itself. At each step the two endpoints live in different components and are therefore non-adjacent, so Lemma 6.3 applies throughout. $\square$

6.4 Examples

Example 6.4 (Single edge). Let $G = K_2$ with edge parameter $\beta \in (0, 1)$ and uniform activity λ . The four configurations give

$$Z_\beta(\lambda) = 1 + 2\beta\lambda + \lambda^2.$$

The two terms $2\beta\lambda$ come from the configurations with exactly one $+1$ spin, both of which cut the edge. The roots of $\lambda^2 + 2\beta\lambda + 1 = 0$ are

$$\lambda = -\beta \pm i\sqrt{1 - \beta^2},$$

which have modulus $\sqrt{\beta^2 + (1 - \beta^2)} = 1$, so both zeros lie on the unit circle at angles $\pi \pm \arccos(\beta)$. As $\beta \rightarrow 0$ they approach $\pm i$; as $\beta \rightarrow 1$ they merge at -1 .

Example 6.5 (Triangle K_3). The all- (-1) and all- $(+1)$ configurations contribute 1 and λ^3 . Each configuration with exactly one $+1$ spin has two cut edges, and by symmetry so does each configuration with two $+1$ spins, giving

$$Z_\beta(\lambda) = \lambda^3 + 3\beta^2\lambda^2 + 3\beta^2\lambda + 1 = (\lambda + 1)(\lambda^2 + (3\beta^2 - 1)\lambda + 1).$$

The quadratic factor has constant term 1, so its roots satisfy $\lambda_1\lambda_2 = 1$, meaning complex roots are conjugates of unit modulus. The discriminant $9\beta^4 - 6\beta^2 - 3$ is negative for all $\beta \in (0, 1)$, so all three zeros lie on the unit circle. The equal coefficients of λ and λ^2 reflect the vertex-transitivity of K_3 .

Example 6.6 (Path P_3). With edges $\{1, 2\}$ and $\{2, 3\}$, enumerating all eight configurations gives

$$Z_\beta(\lambda) = \lambda^3 + (\beta^2 + 2\beta)\lambda^2 + (\beta^2 + 2\beta)\lambda + 1 = (\lambda + 1)(\lambda^2 + (\beta^2 + 2\beta - 1)\lambda + 1).$$

The same argument applies: constant term 1 forces unit modulus on complex roots, and the discriminant is negative on $(0, 1)$. Unlike K_3 , the coefficients of λ and λ^2 are still equal here — this time because P_3 has a reflection symmetry swapping vertices 1 and 3, even though it is not vertex-transitive.

6.5 Stability-Preserving Operators and Lee–Yang Polynomials

The Lee–Yang theorem tells us where the zeros of Z_β cannot be. A natural follow-up question is: which linear operators on polynomial spaces preserve this zero-free property? This is the content of the Borceà–Brändén theory, which we have already used in Section for real stability. Here we state the analogous result for the Lee–Yang setting.

Theorem 6.7 (Borceà–Brändén). *Let $T : \mathbb{C}_\kappa[z_1, \dots, z_n] \rightarrow \mathbb{C}[z_1, \dots, z_n]$ be a non-degenerate linear operator. Then T preserves the Lee–Yang property with respect to the unit disk $\mathbb{D}$ if and only if the polynomial*

$$T \left[\prod_{i=1}^n (1 + z_i w_i)^\kappa \right]$$

is simultaneously $\mathbb{D}^n$ -stable and $(\mathbb{C} \setminus \overline{\mathbb{D}})^n$ -stable.

The condition says that the symbol of T must itself be Lee–Yang. This is a clean analogue of Theorem 4.1: stability-preserving operators are characterised by their symbols, and Lee–Yang-preserving operators are characterised by Lee–Yang symbols.

Example 6.8 (Specialisation preserves Lee–Yang). Fix $a \in \mathbb{C}$ with $|a| = 1$, and let $T_a : f(z_1, z_2) \mapsto f(a, z_2)$. Its symbol is

$$T_a[(1 + z_1 w_1)(1 + z_2 w_2)] = (1 + a w_1)(1 + z_2 w_2).$$

The only root of $1 + a w_1$ is $w_1 = -1/a$, which has modulus 1 since $|a| = 1$. So the symbol is Lee–Yang, and T_a preserves the Lee–Yang property.

Example 6.9 (Asano contraction preserves Lee–Yang). The Asano contraction sets $\lambda_1 = \lambda_2 = \lambda$, giving $T : f(\lambda_1, \lambda_2) \mapsto f(\lambda, \lambda)$. Its symbol is

$$T[(1 + \lambda_1 w_1)(1 + \lambda_2 w_2)] = (1 + \lambda w_1)(1 + \lambda w_2).$$

For any w_1, w_2 on the unit circle, the roots in λ are $-1/w_i$, which also lie on the unit circle. So the symbol is Lee–Yang, confirming Lemma 6.3 as a special case of the general theory.

Example 6.10 (Differentiation need not preserve Lee–Yang). Let $T = \partial/\partial\lambda$ on univariate polynomials. Its symbol is

$$T[(1 + \lambda w)^\kappa] = \kappa w(1 + \lambda w)^{\kappa-1}.$$

For $\kappa = 2$ this gives $2w(1 + \lambda w)$, which has a root at $\lambda = 0$, strictly inside the unit disk. So T does not preserve Lee–Yang: the polynomial $Z_\beta(\lambda) = \lambda^2 + 2\beta\lambda + 1$ is Lee–Yang, but its derivative $Z'_\beta(\lambda) = 2\lambda + 2\beta$ has its root at $\lambda = -\beta$, which satisfies $|\beta| < 1$.

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